259k Parameters to Recognize U: Siamese Alignment of Lensed Fast Radio Bursts

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***Abstract*—When a foreground galaxy lenses a fast radio burst (FRB), the burst reaches Earth twice, days to months apart, and the two copies differ only by propagation effects. Spotting such copies among the bursts of a repeating source is a pairwise verification problem with no real positive examples, adversarial *impostor* negatives that share a propagation imprint, and a required false-positive rate of 1%. We build a semi-synthetic benchmark from all 981 public dynamic spectra of the 83 repeating sources in the CHIME/FRB Catalog 2 and propose a 259k-parameter Siamese network that aligns the feature maps of the two bursts with a differentiable correlation layer before fusing them. In a pre-registered test run once over all 80 eligible sources, each scored by a model that never saw it, the network recovers 0.436 [0.226, 0.675] of lensed pairs at 1% false-positive rate within the simulated nuisance ranges, against 0.211 [0.153, 0.293] for the best physics- or feature-based baseline (paired gain +0.226 [+0.067, +0.403]). Ablations show that both the alignment and impostor training are needed. On 18,585 real cross-source pairs, which are known negatives, the false-positive rate at simulated-twin thresholds is 1.0% [0.5, 1.8]; copies of a real burst are recognized as often as copies of its template; and the pipeline reproduces on a second machine. Survey geometry limits discovery more than the classifier does: a transit telescope records both copies for only about 2% of lenses, whereas an all-sky monitor would find about 50 times more lensed repeaters among the same sources.**

***Index Terms*—Siamese networks, pairwise verification, simulation-to-real transfer, deep learning for signal processing, radio astronomy**

I. Introduction

Fast radio bursts are millisecond radio flashes from distant galaxies [1], [2], and some sources repeat thousands of times. If a galaxy sits close to the line of sight, gravitational lensing splits the source into two images, and every burst then arrives twice, separated by a fixed delay Δt of days to months (Fig. 1a). A single confirmed lensed FRB would measure the expansion rate of the Universe and test dark-matter models [3]–[5], yet none has been found so far [6]–[9].

A search must decide whether two bursts from the same sky position are copies of one burst or two bursts of a repeating source. Three things make this hard. First, there are no real positives, so any detector is trained on simulation, and its transfer to real data must be checked without labeled positives. Second, some negatives are adversarial: distinct bursts of one source can share a spectral envelope imposed by a plasma lens, or a scintillation pattern, and then resemble copies. Third, the chance that a given repeater is strongly lensed is of order 10^{-4} or less (the lensing optical depth is 1.5×10^{-4} at redshift 1 and drops steeply for nearer sources), so decisions must hold at a false-positive rate of 1% or below.

We make four contributions (Fig. 2):

1) a semi-synthetic pair benchmark built from all public CHIME/FRB Catalog 2 repeater spectra, with impostor negatives and frozen, hash-checked evaluation sets;
2) a Siamese correlation network that aligns feature maps before comparing them, with ablations over three seeds;
3) a pre-registered evaluation for a detector without real positives, which combines one test run over all sources, true-negative transfer with per-pair twins, a real-burst stand-in for positives and an independent reproduction;
4) an analysis of where such a detector can be used, which shows that transit telescopes rarely record both copies, so sky coverage limits discovery more than classifier accuracy does.

II. Related Work

Siamese verification. Siamese networks decide whether two inputs share an identity by comparing the outputs of one shared encoder, from signature verification [10] to contrastive embedding [11] and one-shot recognition [12]. Most such networks compare global embedding vectors, which discards where in the input a feature occurs. Patch-matching studies have compared them with two-channel networks that fuse the inputs early [13], and our baselines include both kinds. Correlating feature maps over a range of displacements was introduced for optical flow [14] and later used in Siamese tracking [15]. We use it to estimate and remove the residual time offset between two bursts before comparing them.

Lensed-pair identification. For gravitational waves, lensed event pairs have been identified with a DenseNet and gradient boosting [16], and with a Siamese residual network that fuses features by absolute difference and product [17], which its authors say still needs adapting to real detector data. FRB lensing searches have relied on auto- and cross-correlation statistics [6], [7], [9], and machine learning on FRBs has

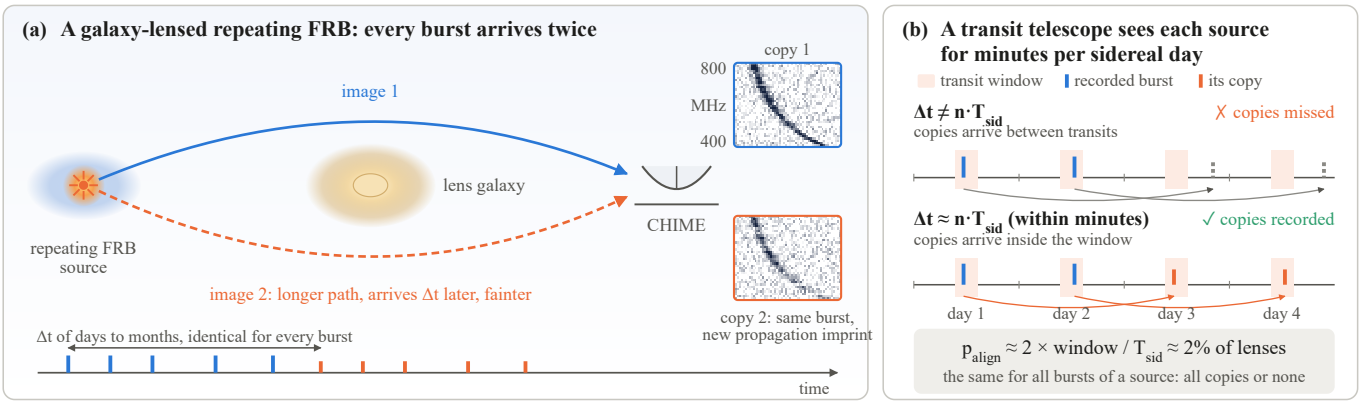


Fig. 1. The problem. (a) A galaxy lens sends every burst of a repeating FRB along two paths, and the copy arrives a fixed Δt later, fainter, with its own scintillation and scattering. The dynamic spectra show one burst sweeping from 800 to 400 MHz (the ν^{-2} dispersion delay), computed from the cold-plasma law. (b) A transit telescope such as CHIME sees a source only within minutes of transit, so it records the copies of recorded bursts only when Δt is close to a whole number of sidereal days.

separated repeating from non-repeating sources [18]. As far as we know, no learned detector of lensed FRB copies exists. Randomizing nuisance parameters makes detectors trained on simulation more robust [19], but their transfer still has to be shown on real data.

III. Problem Formulation

A burst is observed as a dynamic spectrum $X \in \mathbb{R}^{F\times T}$ ($F = 256$ channels, $T = 160$ samples of 0.983 ms, unit off-pulse noise), which we model as an intrinsic burst U seen through one propagation path, $X = \mathcal{P}_\theta(U) + N$: $\mathcal{P}_\theta$ applies the path's time shift, residual dispersion, scattering, scintillation and amplitude, and N is real noise. A pair (A, B) from one sky position is lensed (L) if $U_A = U_B$ with independent θ_A, θ_B, and otherwise a repeat (R, one source), unrelated (D, different sources) or impostor (I, a repeat whose members share a propagation imprint). Finding a lensed pair thus means recognizing the same U behind two differently propagated copies. A detector outputs a symmetric score $s(A, B) = s(B, A)$ and flags a pair when $s > \tau_\alpha$, where τ_α is the $(1-\alpha)$ quantile of s over negatives. Our primary metric is

$$\text{Recall@}\alpha = \Pr[s(A,B) > \tau_\alpha \mid U_A = U_B], \quad \alpha = 1\%, \quad (1)$$

with negatives $\{\text{R}, \text{I}\}$, the alternatives that a real search meets within one source. Because pairs share sources, we report 95% intervals from a bootstrap over sources that recomputes τ_α in every resample. For validation, real data can supply negatives but not a single pair with $U_A = U_B$.

IV. Benchmark Construction

Real inputs. We use the Stokes-I dynamic spectra of all 981 bursts of the 83 repeating sources in CHIME/FRB Catalog 2 [20], recorded at 400 to 800 MHz by the CHIME transit telescope [21]. Each burst is binned to 256×160 around its emission and normalized by its off-pulse noise. The catalog's fitted burst models give noise-free templates (950 are usable), and the off-pulse columns of each file, mirror-tiled, give real noise backgrounds (856), each with its own channel mask.

Pair classes. A lensed pair uses one template twice, a repeat pair two templates of one source, and an unrelated pair templates of two sources. An impostor is a repeat pair in which member B takes on member A's spectral envelope smoothed over 25 MHz (I1, mimicking a plasma lens), in which both members share one scintillation pattern (I2), or both. Each member is injected on a different real background at a measured matched-filter signal-to-noise ratio (S/N), with its own time shift (± 10 samples), residual dispersion, scattering and scintillation, so that only the identity of the burst separates lensed from repeat pairs. Probes that look at one member at a time cannot tell the classes apart (AUC ≤ 0.53).

Splits and frozen sets. Whole sources are assigned to train, validation or test, 60/20/20 by burst count. Validation and test sets of 4000 pairs, and "bright" sets with pair minimum S/N $\geq$ 25 (the pre-registered headline bin), are regenerated from fixed seeds and checked against stored SHA-256 hashes. The test split stayed sealed under the pre-registered protocol until the single test run (Section VI).

V. Siamese Correlation Network

Encoder. Inputs first pass through asinh, which compresses the dynamic range of bright bursts. A shared encoder g (Fig. 3) of four 5×3 convolution blocks with group normalization maps X to $g(X) \in \mathbb{R}^{C\times F'\times T'}$ with $C = 64$, $F' = 16$ and $T' = 80$. We keep two-sample time resolution because lensed copies differ by residual time offsets.

Correlation and soft alignment. For lags $\ell \in \{-L, \ldots, L\}$ with $L = 12$ encoder steps (± 24 samples),

$$\rho(\ell) = \frac{\langle \bar{g}_A,\ \bar{g}_B(\ell)\rangle}{\|\bar{g}_A\|\,\|\bar{g}_B(\ell)\|}, \qquad w_\ell = \frac{e^{\beta\rho(\ell)}}{\sum_k e^{\beta\rho(k)}}, \quad (2)$$

where $g_B(\ell)$ is g_B shifted by ℓ in time, the bar denotes removal of the time mean, and β is learned. The aligned map $\tilde{g}_B =$

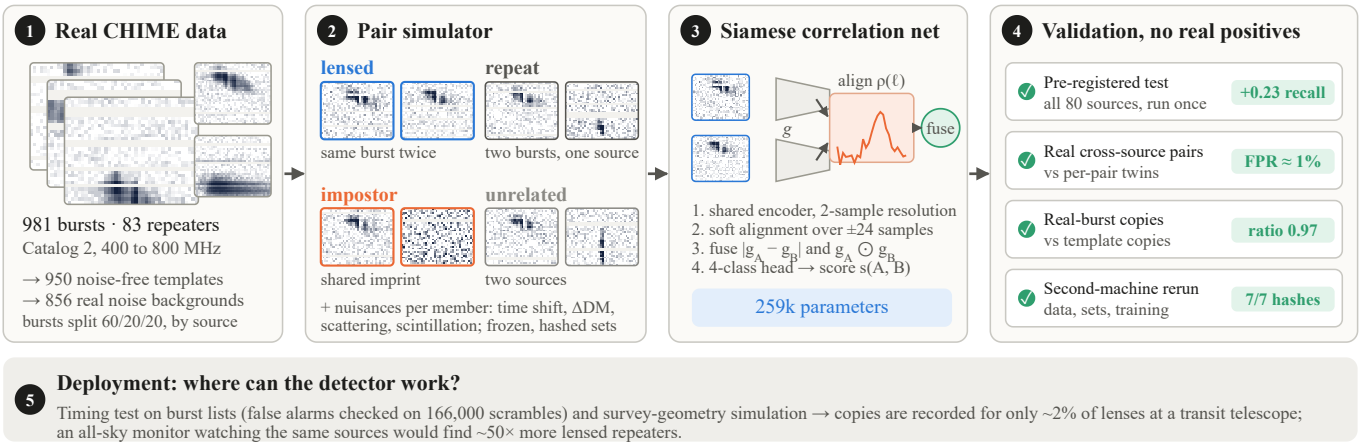


Fig. 2. Workflow. (1) Real CHIME/FRB Catalog 2 dynamic spectra (tiles show real bright bursts) provide noise-free templates and real noise backgrounds. (2) The pair simulator places templates on different real backgrounds with independent nuisances to make four classes; impostors share a spectral envelope or a scintillation pattern. (3) The Siamese correlation network scores each pair. (4) Validation uses only real negatives, a real-burst proxy and an independent rerun. (5) A survey-geometry analysis decides where the detector can be deployed.

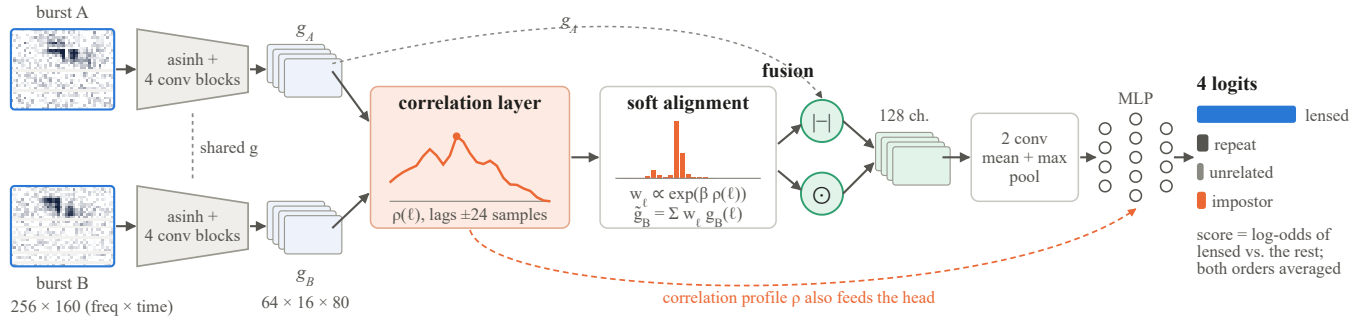


Fig. 3. Model 6. A weight-shared encoder maps each burst to a $64 \times 16 \times 80$ feature map at two-sample time resolution. The correlation layer measures the normalized correlation $\rho(\ell)$ over lags of ± 24 samples (the curve is computed for the pair shown), softmax weights align g_B to g_A, and the aligned maps are fused by absolute difference and product. The correlation profile also feeds the head. Both input orders are computed and averaged.

$\sum_\ell w_\ell\, g_B(\ell)$ is a differentiable estimate of the residual delay, which the network can sharpen or soften through β.

Fusion and head. The fused map $[\,|g_A - \tilde{g}_B|,\ g_A \odot \tilde{g}_B\,]$ passes through two convolution blocks and then mean and max pooling. An MLP takes the pooled features together with ρ and outputs four logits z. The lensed score is the log-odds of the lensed class against the others, averaged over both input orders to make it symmetric:

$$s(A,B) = \tfrac{1}{2}\left[\sigma(A,B) + \sigma(B,A)\right], \quad \sigma = z_{\mathrm{L}} - \log \sum_{c \neq \mathrm{L}} e^{z_c}. \tag{3}$$

The model has 259k parameters.

Training. All learned models are trained for 20,000 steps on batches of 64 pairs drawn on the fly from training sources (half with an S/N floor of 10, half with 25), using four-class cross-entropy, AdamW (learning rate 10^{-3}, weight decay 10^{-4}), 1000 warm-up steps and cosine decay. The checkpoint is selected on the validation bright set.

VI. Evaluation Without Real Positives

Pre-registered test. Every choice described here was written into a plan before the result it affects was known. The primary comparison is model 6 against the best of the cross-correlation, physics and feature baselines (Section VII) on the validation set, which is the GBM. Model 6 counts as better only if the lower 95% bound of the paired recall difference is above zero. The validation interval was wide (half-width 0.16), so a pre-registered trigger replaced the fixed test split by five source-grouped folds over all 80 sources with at least two templates (16 per fold). Settings were frozen, with one seed per fold and the final checkpoint, so every source is scored by a model that never saw it. A rehearsal on the 60 development sources gave +0.22 [+0.04, +0.38]. The bootstrap pools one threshold across folds, which penalizes differences in score scale between fold models. We opened the test sources once, after all methods had been frozen at a version tag. One deviation must be reported: the test was run before the nuisance ranges of the propagation model were formally signed off. Its result reached the team's shared repository only

after the sign-off, whose record still lists the test as unopened, and the sign-off used validation pairs only and left the ranges, the frozen sets and all methods unchanged; a wide-scattering training variant was rejected by a rule fixed before it was run. We therefore treat the result as the pre-registered comparison. The fixed-split test sets (20 sources) serve as secondary checks.

True-negative transfer. Two bursts from different sources cannot be lensed copies, so real cross-source pairs are known negatives that we can score without opening the search on same-source pairs. For each real pair we simulate five *twins* from the same two templates at their measured S/N and put the threshold at the 99th percentile of the twins. The pre-registered criterion is a real false-positive rate of at most 2% in each S/N bin. Matching the twins to the pairs matters. Against generic simulated pairs, cross-correlation seems to fail below S/N 25 (6.2, 3.8, 2.4%), but its own twins fail just as often (5.6, 4.0, 2.5%) because those real pairs are brighter than generic ones, so the apparent failure comes from sample composition.

Real-burst positives. As a stand-in for lensed pairs, we inject two copies of a real burst's cutout, which keeps its fine structure and real noise, with independent nuisances on different backgrounds. Template copies and noise-only copies are drawn alongside with the same random numbers.

Reproduction. A second machine downloaded and processed all 981 spectra again, rebuilt the injection cache, regenerated the frozen sets, retrained model 6 with three seeds and repeated the checks.

VII. Experiments

Baselines. All methods score the same pairs and average over both member orders: (1) normalized cross-correlation in the on-pulse box, maximized over ± 24 lags, with 16 channel blocks; (2) a physics statistic that fits one member to the other over scattering time, residual dispersion, lag and amplitude; (3) gradient-boosted trees (GBM) on seven measured burst properties per member, the best of the first three and the pre-registered comparator; (4) a two-channel CNN; (5) a classic Siamese network that compares embedding vectors; and (7) cross-attention between time-frequency patches.

Main result. Model 6 meets the pre-registered criterion. Over all 80 sources it recovers 0.436 [0.226, 0.675] of lensed pairs against 0.211 [0.153, 0.293] for the GBM, a paired gain of +0.226 [+0.067, +0.403] whose lower bound is above zero (Table I). With a separate threshold for each fold the recalls would be 0.506 and 0.226, so the pooled threshold costs model 6 more than the GBM. The gain agrees with the development rehearsal (+0.22 [+0.04, +0.38]) and appears on every set: +0.39 on validation and +0.323 [+0.115, +0.543] on the 20 test sources, where the second machine's seeds stand in for model 6. On validation, every learned model that trained scores above all three baselines. The AUCs are much closer (0.982 against 0.936), which places the gain in the low-false-positive tail where a search operates. Recall varies between folds (0.19 to 0.86 for model 6, 0.16 to 0.29 for the GBM), a spread that the source bootstrap carries into the interval.

TABLE I

Recall at 1% FPR (lensed vs repeat + impostor), S/N $\geq$ 25, 95% source bootstrap. Validation: development set, used to select checkpoints. Test sources: sealed until the single test run. All 80 sources: the pre-registered primary, out-of-fold (OOF) over five source-grouped folds, final checkpoints. Empty cells were not evaluated.

Method	Validation (26 sources)	Test sources (20 sources)	All 80 sources (grouped-fold OOF)
(1) Cross-correlation	0.171	0.128	
(2) Physics fit	0.164	0.070	
(3) Feature GBM	0.285 [0.180, 0.401]	0.226 [0.152, 0.390]	0.211 [0.153, 0.293]
(4) Two-channel CNN	0.423 / 0.409 / 0.407		0.299 [0.218, 0.474]
(5) Classic Siamese	0.500 / 0.559 / 0.541		0.368 [0.190, 0.566]
(7) Cross-attention	0.471 / 0.013[a] / 0.579		0.264 [0.156, 0.411]
(6) Ours	**0.701 / 0.698 / 0.684**	**0.596 / 0.459 / 0.594**[b]	**0.436 [0.226, 0.675]**
(6) − (3), paired	+0.394 [+0.226, +0.542]	+0.323 [+0.115, +0.543]	**+0.226 [+0.067, +0.403]**
AUC, (6) / (3)	0.987 / 0.925	0.976 to 0.985 / 0.934	0.982 / 0.936

[a] One of three seeds collapsed to a plateau. [b] Model 6 seeds retrained on a second machine, whose validation recall is 0.05 lower (0.668 / 0.649 / 0.616), so this column is conservative for model 6.

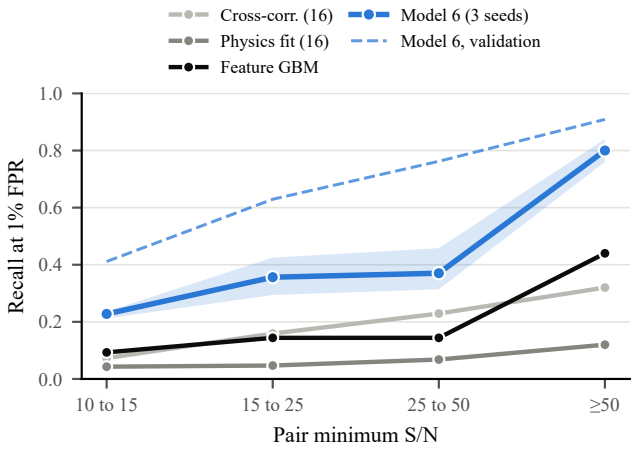


Fig. 4. Recall at 1% FPR against pair minimum S/N on the test set (20 test sources; 537, 320, 118 and 25 lensed pairs per bin), from the single test run. Model 6: mean and range of the three second-machine seeds; dashed: the same seeds on the validation set.

Sensitivity and errors. On the test sources the gain holds in every S/N bin (Fig. 4). Every method recalls less than on validation (Table I), the untrained cross-correlation and physics fit included, which suggests that the test sources are intrinsically harder. At the pooled 1% threshold of the primary test, 191 of 200 false positives of model 6 are impostors with a shared envelope (I1 or I1+I2) and only 7 are plain repeats, so the remaining errors sit in the physically hardest case. Model 6 (seed 0) labels 95% of lensed pairs from unseen sources correctly but only 32% of their repeat pairs, compared with 86% on training sources. It mostly calls the unseen repeats unrelated, so part of its capacity memorizes source identity. Three training variants aimed at this (merging the repeat and unrelated targets, a random bandpass per member, or both) changed recall by -0.02 to +0.00, with every interval including zero.

Ablations. Removing the soft alignment, or training without impostors, lowers recall in every seed (Table II). Without im-

TABLE II

ABLATIONS OF MODEL 6 ON THE SECOND MACHINE, THREE SEEDS EACH: SEED-MEAN RECALL AT 1% FPR AND PAIRED DIFFERENCE FROM THE FULL MODEL (VALIDATION BRIGHT SET, 95% SOURCE BOOTSTRAP).

Variant	Recall	Difference
Full model	0.644	
No soft alignment	0.557	−0.087 [−0.186, −0.025]
No impostors in training	0.474	−0.170 [−0.254, −0.052]
Hard-negative repeats	0.698	+0.053 [−0.011, +0.082]

TABLE III

REAL-DATA VALIDATION OF MODEL 6: VALIDATION SOURCES ON TWO MACHINES, AND ALL 80 SOURCES WITH THE OUT-OF-FOLD MODELS OF THE TEST RUN (95% SOURCE BOOTSTRAP WHERE GIVEN). EMPTY CELLS WERE NOT EVALUATED.

	Validation sources		All 80 sources
Check (criterion)	Original	Second machine	(OOF models)
Real FPR at twin threshold, S/N ≥ 25 (≤ 2%)	1.1 / 1.4 / 0.8%	1.5 / 1.3 / 1.0%	1.0% [0.5, 1.8]
GBM, same check	0.9%	1.0%	0.9%
Real-to-twin rank correlation	0.91 to 0.94	0.91 to 0.92	0.91
Twin-to-twin ceiling	0.88 to 0.91	0.87 to 0.89	0.90
Real-burst / template recall, S/N 25 (≥ 0.8)	0.96 [0.89, 1.06]	0.97 [0.91, 1.08]	
Real-burst / template recall, S/N 40	0.99 [0.91, 1.13]	1.04 [0.97, 1.18]	
Noise-only copies flagged (≤ 0.05)	0.000	0.000	
Frozen-set hashes	reference	7/7 match	

postors in training, the network takes envelope-sharing bursts of one source for copies. Oversampling repeats with similar templates helps slightly in every seed, but its interval includes zero, and on a closed-form benchmark the same scheme biased a learned likelihood ratio low by about 1 dex, so we do not adopt it without a correction. Among the architectures on the 80-source folds, model 6 leads the classic Siamese network by +0.068 [−0.013, +0.154], the two-channel CNN by +0.124 [−0.027, +0.287] and cross-attention by +0.172 [+0.058, +0.307]. Only the last interval excludes zero, so the folds support the ranking without resolving every gap. The CNN and the classic Siamese network also beat the GBM (+0.102 [+0.024, +0.213] and +0.158 [+0.031, +0.286]), while the interval for cross-attention includes zero (+0.054 [−0.011, +0.138]).

Real-data validation. On the 3851 real cross-source pairs of the validation sources, all 12 trained models and 3 baselines meet the twin-threshold criterion once short real windows are filled out with off-pulse noise, as the simulated backgrounds already are. Model 6's real-to-twin rank correlation also reaches the twin-to-twin ceiling on both machines (Table III). Before the fill, model 6 flagged real pairs whose windows shared zero-padded edges, a format mismatch that the twin test exposed. Two copies of a real burst, with all its fine structure, are recognized as often as two copies of its smooth template, and noise-only copies never are. Model 6 therefore does not rely on template smoothness or shared noise, whereas the classic Siamese network does (ratio 0.72 to 0.84). In the test run, with each fold's model scoring real pairs of its own held-out sources, the check passes for model 6 and all three baselines on all 80 sources (18,585 pairs, at most 4000 per fold). Thresholds frozen on validation, by contrast, give a realized false-positive rate of 1.5 to 3.4% for model 6 on the test sources (GBM: 1.5%), so a search must set its thresholds on real negatives.

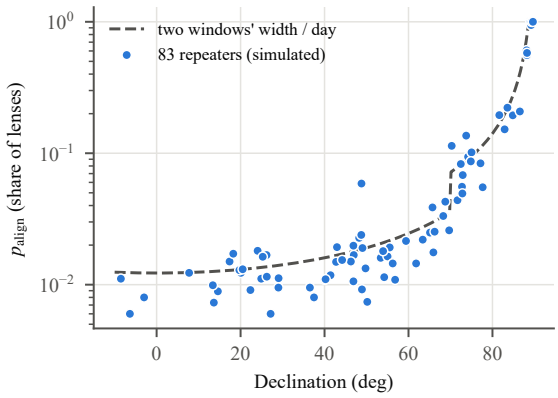


Fig. 5. Fraction of lens delays for which CHIME records the copies of a repeater's bursts, for the 83 Catalog 2 repeaters (simulated with their own burst times, exposure and redshift), with the approximation (4) dashed.

Efficiency. Model 6 scores 1149 pairs per second, in both orders, on one RTX 5070 Ti (bfloat16), so all 78,701 same-source pairs of Catalog 2 take about 70 s.

VIII. SURVEY GEOMETRY: WHERE CAN THE DETECTOR WORK?

A lensed repeater can also be found without a classifier. A galaxy lens fixes Δt far more precisely than the catalog's 4 ms timing, so two burst pairs (i, j) and (k, l) of one source with $|\Delta t_{ij} - \Delta t_{kl}|$ below four combined timing errors already form a significant pattern. We calibrated this timing test on burst lists scrambled by whole sidereal days, which preserves each burst's hour angle, and confirmed its false-alarm rate on independent scrambles (1 of 166,000 held-out scrambles).

Both the classifier and the timing test need the copies to be recorded. CHIME is a transit telescope, so every recorded burst arrives within a window of width $W(\delta) = 2\theta_{\rm EW}/(15^\circ\,{\rm h}^{-1}\cos\delta)$ around transit, with $\theta_{\rm EW} \approx 1.1^\circ$ measured from the real bursts. The copy of each burst arrives at the same hour-angle offset, $24\,{\rm h} \times {\rm frac}(\Delta t/T_{\rm sid})$, so the copies of either all bursts or none are recorded (Fig. 1b). For a broad delay distribution

$$p_{\rm align} \approx 2W(\delta)/T_{\rm sid}, \tag{4}$$

doubled above $\delta = 70^\circ$, where the lower transit adds a second window. Simulating singular-isothermal-sphere lenses with each source's redshift, burst times and exposure gives a median $p_{\rm align}$ of 1.8%, in line with (4) (Fig. 5). Repeating FRBs therefore face the transit problem already known for one-off bursts [22].

The expected number of detectable lensed repeaters in the Catalog 2 sample is 5×10^{-6} without magnification bias and 1×10^{-4} with it (slope 1.5), quoted to one significant figure because independent Monte Carlo reruns differ by tens of percent. A null search is the expected outcome. A 10-year transit survey gains less than a factor of two, because

the alignment requirement remains, whereas an all-sky monitor watching the same sources gains a factor of about 50. Raising the median host-galaxy dispersion measure from 68 to $150\,\mathrm{pc\,cm^{-3}}$ changes the yield by about 20%, within this scatter. In transit surveys, including next-generation drift-scan arrays such as CHORD [23], the classifier would vet the rare single recorded copy and reject impostors. Its full value lies with wide-field monitors such as BURSTT [24].

IX. Discussion and Limitations

Simulated false-positive rates are useful for ranking methods, but deployment thresholds have to come from real negatives. Twin matching makes such a check interpretable: it separates simulation gaps from sample composition, and here it exposed a format mismatch that no simulated benchmark would reveal. Pre-registration gives the single test run its weight: every rule was fixed before the test sources were seen, and the gain on them matches the development estimate. Several limitations remain. The test was run before the propagation nuisance ranges were formally signed off, and a later sweep on validation pairs shows that scattering is the range that matters. Recall is unchanged up to a differential scattering time between the copies of $\tau_{600} = 0.3\,\mathrm{ms}$, but it falls to 0.57 of its value at 1 ms, the top of the training range, and to 0.13 at 3 ms. Extending training to 10 ms raises the 3 ms value only to 0.24. The physics fit, which models scattering, keeps its recall up to 3 ms, so the loss is a limitation of shape-matching models. Images seen through a scattering disc in the lens galaxy [4] would be missed, and the detector suits lenses with little cold gas, which dominate the lensing optical depth. Because the positive class is simulated, absolute recall depends on the impostor model and on how faithfully the templates represent bursts. The real-burst proxy and the transfer checks bound this dependence but do not remove it. Twenty sources per test set and 16 per fold leave wide intervals. The second machine reproduced the data bit for bit but placed model 6 about 0.05 recall below the original seeds, for reasons not yet isolated. Finally, all lenses are modeled as singular isothermal spheres.

X. Conclusion

A small Siamese network that aligns two bursts before comparing them identifies lensed copies of real FRB spectra about twice as often as physics and feature baselines at a 1% false-positive rate, within the simulated nuisance ranges, in a pre-registered test run once over all sources. It passes real-data checks designed for a detector without real positives, and it reproduces on a second machine. The same analysis shows that transit geometry is a bigger obstacle to finding lensed repeaters than classification accuracy, which points detectors of this kind toward wide-field radio arrays. All inputs are public (CHIME/FRB Catalog 2 [20]), and the code, frozen-set manifests, hashes and pre-registered plans are available from the authors on request.

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